PROJECT PROBLEM STATEMENT & CHARTER

EventHub: Campus & Community Event Management Platform

A Full-Stack Event Orchestration, Digital Ticketing, and Concurrency-Safe Reservation System

Module:	Web & Mobile Application Technologies (SE2020)	**Institution:**	Sri Lanka Institute of Information Technology (SLIIT)
Target Architecture:	React Native / Expo Web + Node.js + MongoDB Atlas	**Document Status:**	Final Academic Release (v1.0)
Author:	lakshan (SE2020 Student Developer)	**Academic Year:**	2026/2027

1. Executive Summary

In contemporary tertiary academic institutions and community ecosystems, extracurricular, professional, and technical events serve as crucial catalysts for student growth and collaborative networking. However, the logistical coordination of campus events remains plagued by decentralized communication channels, manual ticket issuance, and fragmented administrative oversight. **EventHub** is an engineering response to these systemic inefficiencies. By combining an intuitive React Native mobile and web client with a secure, concurrency-controlled RESTful API powered by Node.js, Express, and MongoDB Atlas, EventHub consolidates event discovery, real-time capacity allocation, and digital ticketing into a single, unified enterprise ecosystem.

2. Background & Industry Context

Educational campuses regularly host conferences, hackathons, guest lectures, cultural concerts, and academic workshops. Conventionally, organizers rely on disparate tools: social media channels for announcements, generic survey forms for participant registration, and manual physical check-ins on paper rosters. While seemingly simple, this disjointed workflow fails rapidly when participant interest scales beyond a few dozen individuals. There is no automated synchronization between the number of available physical seats and inbound booking requests, resulting in administrative chaos, venue fire-code violations, and severe student frustration.

3. Detailed Problem Statement

Empirical analysis of existing campus event workflows reveals four principal failure points that hinder operational effectiveness and degrade user experience:

3.1 Information Silos & Fragmented Discovery

- **Lack of a Centralized Repository:** Event notices are scattered across social media feeds, chat groups, and physical corkboards. Prospective attendees frequently miss registration deadlines or fail to discover events relevant to their career goals.
- **Static and Non-Interactive Catalogs:** Existing promotional mechanisms do not provide category filtering, search capabilities, or dynamic calendar timeline alignment, creating high discovery friction.

3.2 Inventory Race Conditions & Overbooking Hazards

- **Uncontrolled Concurrent Submissions:** When high-demand events open registration, hundreds of users apply simultaneously. Without strict atomic inventory counters, registration forms continue accepting submissions long after venue capacity is exceeded.
- **Physical Safety & Venue Compliance:** Overbooked venues force organizers to either deny entry to registered attendees at the door or breach university occupational safety regulations by packing lecture halls beyond capacity.

3.3 Paper-Based Inefficiencies & Fraud Susceptibility

- **High Ecological & Monetary Cost:** Printing hundreds of paper ticket stubs or wristbands consumes financial resources and generates unneeded paper waste for short-duration events.
- **Counterfeit Vulnerabilities:** Paper passes lack cryptographic validation or digital tracking, making them prone to duplication, loss, or unauthorized transfer without organizer knowledge.

3.4 Absence of Self-Service Cancellation & Capacity Recirculation

- **Ghost Bookings:** A significant percentage of attendees who register are subsequently unable to attend. In existing systems, there is no self-service mechanism for attendees to cancel a booking.
- **Permanent Capacity Lock:** Unattended tickets remain permanently allocated in spreadsheets, blocking motivated attendees on waiting lists from claiming valid vacant seats.

4. The Proposed Solution: EventHub Architecture

EventHub addresses each identified challenge through an integrated, cloud-native software architecture:

- **1. Unified Event Discovery Portal:** A sleek, category-driven mobile and web interface enabling instantaneous search and filtering across Tech, Music, Workshops, Sports, and Business events.
- **2. Atomic Seat Decrement Logic:** An ACID-compliant reservation engine that dynamically checks tickets ≤ availableSeats and decrements seat inventory atomically during transactions, guaranteeing 0% overbooking.
- **3. Automated Seat Restoration:** One-tap booking cancellations that instantaneously restore reserved seats to the public inventory, eliminating ghost bookings.
- **4. Digital Ticket Pass Generation:** Generation of interactive, authenticated digital passes containing booking reference IDs, user credentials, and event metadata directly within the attendee's personal wallet.
- **5. Creator Authorization Engine:** Granular role-based controls allowing event creators to modify event parameters or safely cancel events while preventing unauthorized administrative tampering.

5. Project Objectives (SMART Criteria)

Objective	Target Metric	SMART Alignment
Eliminate Overbooking	0% overbooking rate across all concurrent booking operations	Specific & Measurable through server-side capacity checks and atomic reservation locks.
Sub-Second Response	API response time < 500ms for event browsing and bookings	Achievable via optimized MongoDB indexing and lean JSON payloads.
Paperless Operations	100% digital pass delivery on mobile/web profiles	Relevant to modern sustainability standards and instant user gratification.
Self-Service Workflow	< 5 seconds to cancel and automatically restore seat capacity	Time-bound, automated database updates on state changes.
Cross-Platform Reach	Single unified codebase running on Android, iOS, and Web	Realized through React Native and Expo web bundler architecture.

6. Stakeholder & Target Audience Analysis

The system serves three primary user personas, each with distinct interaction patterns:

- **Student Attendees (Consumers):** Seek effortless event discovery, transparent seat availability counts, frictionless 1-tap bookings, and easy access to digital ticket passes without printing requirements.
- **Event Organizers (Producers):** Require an intuitive authoring dashboard to publish events with banner artwork, establish seat quotas, monitor real-time ticket sales, and manage event schedules.
- **Campus Administration (Auditors):** Need compliance assurance that event capacities do not exceed physical hall limits and that student engagement records are secure and verifiable.

7. Scope of the Implementation

Functional Module	In-Scope Capabilities (Implemented)	Future Roadmap (Out-of-Scope)
Identity Management	JWT Token issuance, Bcrypt password hashing (10 salt rounds), persistent AsyncStorage session.	OAuth2 Single Sign-On (Google/SLIIT MS Account), biometric fingerprint login.
Event Management	Full CRUD operations, Multer multipart image uploads, category classification, creator authorization checks.	Recurring multi-day schedules, multi-speaker agenda tables.
Booking & Inventory	Seat availability checking, atomic decrement logic, automatic capacity restoration upon cancellation.	Paid payment gateway (Stripe/PayHere), tiered VIP/standard seating tiers.
Verification	Unique MongoDB ObjectId ticket identifiers, real-time status badges (Confirmed, Cancelled).	Camera-based QR code ticket scanning at physical hall entrances.

8. Technical Architecture & Feasibility

The project demonstrates technical feasibility through a validated, enterprise-grade open-source technology stack:

- **Frontend Layer:** Built with React Native and Expo SDK 57, enabling universal deployment across native Android/iOS and responsive Web. Features customized HSL tokens, smooth micro-interactions, and secure AsyncStorage tokens.
- **Backend Layer:** Node.js runtime with Express.js routing engine. Employs modular architecture (Controllers, Middleware, Models, Routes) with centralized error interceptors.
- **Persistence Layer:** MongoDB Atlas cloud database with Mongoose ODM. Implements referenced schemas with relational population (Event.createdBy → User, Booking.eventId → Event).
- **Deployment:** Continuous deployment pipeline hosted on Render cloud infrastructure, serving both the RESTful API endpoints and the compiled single-page application bundle.

9. Expected Impact & Value Realization

The deployment of EventHub transforms campus event administration from a reactive, labor-intensive manual task into a proactive, autonomous digital experience. Organizers achieve 100% capacity control without manual spreadsheet reconciliation, while students enjoy instant, paperless reservations directly on their mobile devices. In academic evaluations, the platform validates mastery of full-stack engineering, asynchronous API protocols, state synchronization, and database integrity.

Prepared By: lakshan	**Verified By:** SLIIT Academic Supervisor
Signature: _____	**Signature:** _____
Date: October 2026	**Status:** Approved for Implementation